CSA1110 - OBJECT ORIENTED ANALYSIS & DESIGN

CO4 - ASSESSMENT TOOL-1 - Class test

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1) Architectural Pattern AND MVC.

⇒ An architectural pattern is a reusable structure for organizing compounds of an object-oriented system.

→ Two common architectural patterns:

Model-View-Controller (MVC)

Layered Architecture

MVC consist of:

Mode - manages data and business information.

View - displays information to the user controller

- handles user request

Benefits:

MVC provides separation of concern, improves modularity, maintainability, reusability and testing.

2.) DAO PATTERN:

DAO (Data Access Object) is a design pattern that separates database operations from application logic.

TWO ADVANTAGES:

1.) "Separation of concerns.

2.) Easier maintenance".

WORKING:

Application Logic → DAO → Database.

The DAO handles operations such as insert, update, delete, and retrieve, which application logic focuses on the required functionality.

OBJECT-ORIENTED DESIGN GUIDELINES & LSP.

Object-oriented design guidelines are principles used to create software that is reusable, maintainable and flexible.

Two categories:

1.) Class/object, design guideliness.

2.) Relationship / Inheritance design guideliness.

Liskov substitution principle (LSP):

objects of a superclass should be usable wherever objects of the superclass are expected without affecting program correctness.

importance: LSP improves reusability, polymorphism, maintainability and reliability.

DATA BASE PERSISTENCE AND ORM.

Database persistence means storing data permanently so that it remains available after the application ends.

→ TWO persistence mechanisms:

1) ORM 2) Direct database access using SQL

ORM (object-Relational Mapping) map:

- class → Table.
- object → Row
- Attribute → column

ORM helps store, retrieve, update and delete object in a database with less manual SQL.

5) APPLICATION LAYER.

The application layer coordinates application specific tasks and use cases.

→ Two Responsibilities.

1.) Coordinate use cases.

2.) Manage application workflow.

WORKING:

presentation layer → Application Layer → Domain Layer

→ The Application Layer coordinates the required operations do not implement domain business rule.

→ Business rules are handled by Domain / Business Layer.

→ The separation improves maintainability and clear responsibility.